```
!pip install kaggle
```

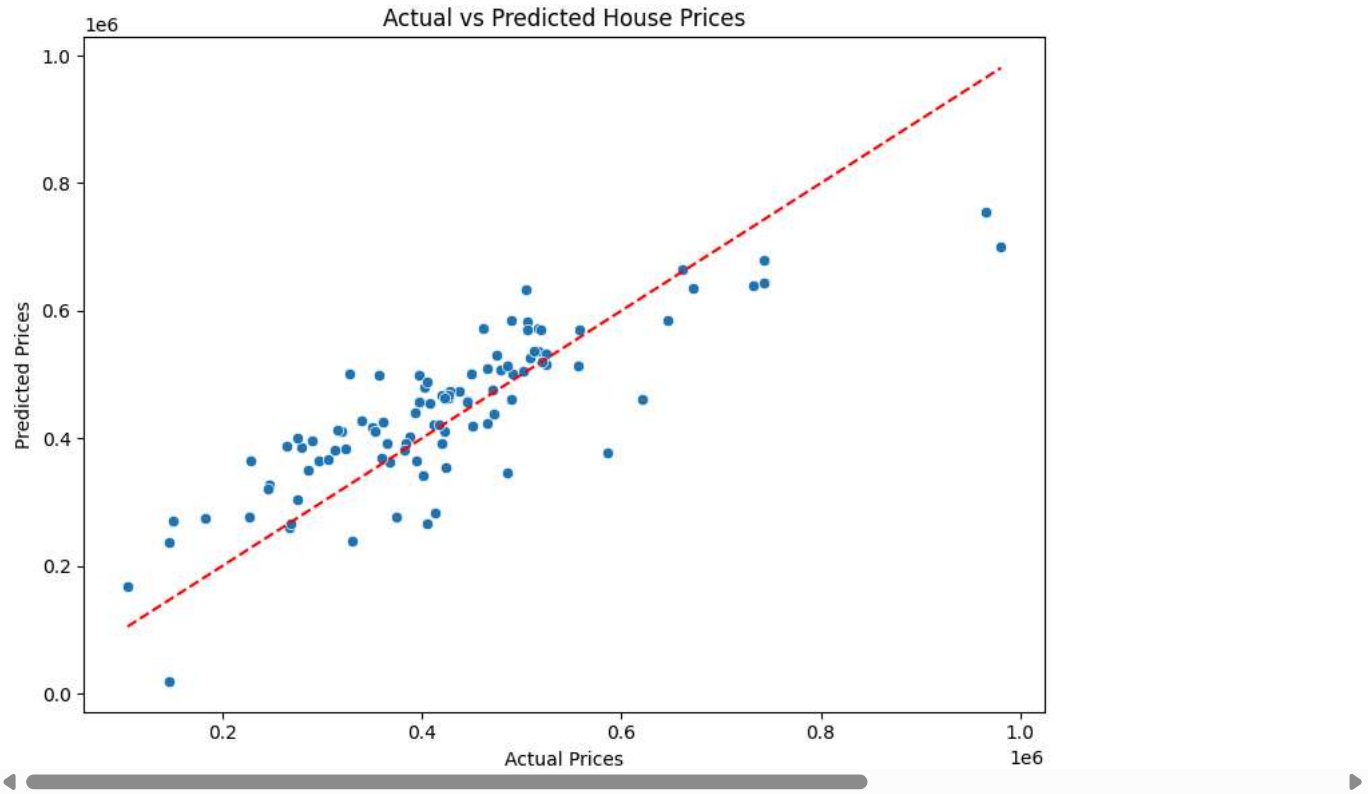
```
Requirement already satisfied: kaggle in /usr/local/lib/python3.11/dist-packages (1.7.4.2)
Requirement already satisfied: bleach in /usr/local/lib/python3.11/dist-packages (from kaggle) (6.2.0)
Requirement already satisfied: certifi>=14.05.14 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2025.4.26)
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.11/dist-packages (from kaggle) (3.4.1)
Requirement already satisfied: idna in /usr/local/lib/python3.11/dist-packages (from kaggle) (3.10)
Requirement already satisfied: protobuf in /usr/local/lib/python3.11/dist-packages (from kaggle) (5.29.4)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.11/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.32.3)
Requirement already satisfied: setuptools>=21.0.0 in /usr/local/lib/python3.11/dist-packages (from kaggle) (75.2.0)
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.11/dist-packages (from kaggle) (1.17.0)
Requirement already satisfied: text-unidecode in /usr/local/lib/python3.11/dist-packages (from kaggle) (1.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from kaggle) (4.67.1)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.4.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.11/dist-packages (from kaggle) (0.5.1)
```

New Section

```
# 1. Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 2. Load your dataset

df = pd.read_csv('/content/housing.csv')

# 3. Explore the dataset
print(df.head())
print(df.info())
print(df.describe())

# 4. Preprocessing (basic example — modify based on your dataset columns)

# Drop rows with missing values
df.dropna(inplace=True)

# Optional: encode categorical variables if any
# Example: If there's a 'location' column
if 'location' in df.columns:
    df = pd.get_dummies(df, columns=['location'], drop_first=True)

# 5. Feature selection — update these based on actual column names in your CSV
print(df.columns) # Print the dataframe's columns to identify the correct names
# Fixing the indentation for these lines:
feature_cols = ['RM', 'LSTAT', 'PTRATIO']  # Example: Replace with your actual feature column names
X = df[feature_cols]


# 6. Train/Test split
# Assuming 'y' corresponds to the target variable (e.g., 'MEDV' for housing prices)
# Make sure 'y' is defined before this line:
y = df['MEDV']  # Replace 'MEDV' with your actual target column name if different
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 7. Train the Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# 8. Make predictions
y_pred = model.predict(X_test)

# 9. Evaluate the model
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print(f"\nModel Evaluation:")
```

```
print(f"\nModel Evaluation: ")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")
print(f"R² Score: {r2:.2f}")

# 10. Visualize predictions vs actual
plt.figure(figsize=(8,6))
sns.scatterplot(x=y_test, y=y_pred)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')  # ideal line
plt.tight_layout()
plt.show()
```

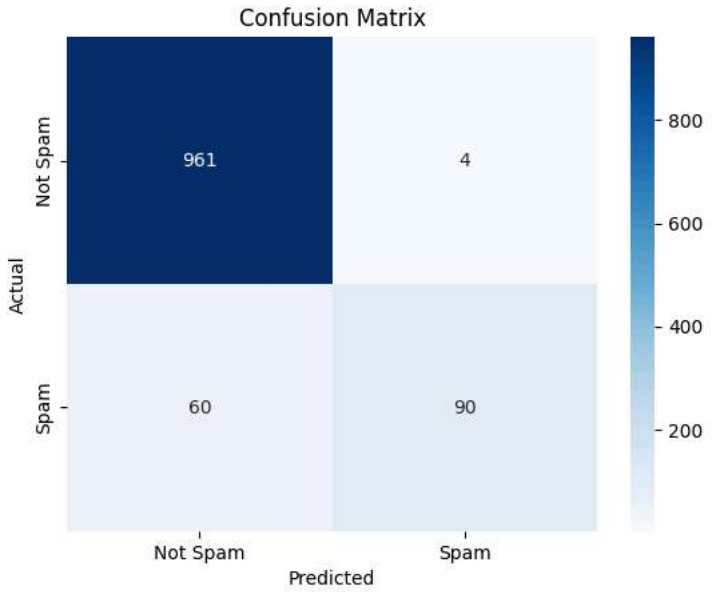
```
      RM  LSTAT  PTRATIO      MEDV
0  6.575   4.98     15.3  504000.0
1  6.421   9.14     17.8  453600.0
2  7.185   4.03     17.8  728700.0
3  6.998   2.94     18.7  701400.0
4  7.147   5.33     18.7  760200.0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 489 entries, 0 to 488
Data columns (total 4 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   RM       489 non-null    float64
 1   LSTAT    489 non-null    float64
 2   PTRATIO  489 non-null    float64
 3   MEDV     489 non-null    float64
dtypes: float64(4)
memory usage: 15.4 KB
None
                RM       LSTAT     PTRATIO          MEDV
count  489.000000  489.000000  489.000000  4.890000e+02
mean     6.240288   12.939632   18.516564  4.543429e+05
std      0.643650    7.081990    2.111268  1.653403e+05
min      3.561000    1.980000   12.600000  1.050000e+05
25%      5.880000    7.370000   17.400000  3.507000e+05
50%      6.185000   11.690000   19.100000  4.389000e+05
75%      6.575000   17.120000   20.200000  5.187000e+05
max      8.398000   37.970000   22.000000  1.024800e+06
Index(['RM', 'LSTAT', 'PTRATIO', 'MEDV'], dtype='object')

Model Evaluation:
Root Mean Squared Error (RMSE): 82395.54
R² Score: 0.69
```




```
# 1. Import Libraries
import pandas as pd
```

```
import numpy as np
import string
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix, classification_report
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import LabelEncoder

import nltk
from nltk.corpus import stopwords
nltk.download('stopwords')

# 2. Load Dataset (Example: 'spam.csv' with 'text' and 'label' columns)
df = pd.read_csv("/content/spam.csv", encoding='latin-1')[['v1', 'v2']]
df.columns = ['label', 'text']
df['label'] = df['label'].replace('ham', 'not spam')

# 3. Preprocessing
def clean_text(text):
    text = text.lower()
    text = ''.join([char for char in text if char not in string.punctuation])
    tokens = text.split()
    tokens = [word for word in tokens if word not in stopwords.words('english')]
    return ' '.join(tokens)

df['clean_text'] = df['text'].apply(clean_text)
df['label_num'] = LabelEncoder().fit_transform(df['label'])

# 4. Feature Extraction
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(df['clean_text'])
y = df['label_num']

# 5. Train/Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 6. Build and Train Logistic Regression Model
model = LogisticRegression()
model.fit(X_train, y_train)

# 7. Evaluation
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Confusion matrix
conf_mat = confusion_matrix(y_test, y_pred)
sns.heatmap(conf_mat, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Not Spam', 'Spam'], yticklabels=['Not Spam', 'Spam'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

# 8. (Optional) Threshold Tuning
y_probs = model.predict_proba(X_test)[:, 1]
custom_threshold = 0.4
y_custom = (y_probs >= custom_threshold).astype(int)

print(f"\nAfter threshold adjustment to {custom_threshold}:")
print("Precision:", precision_score(y_test, y_custom))
print("Recall:", recall_score(y_test, y_custom))
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
Accuracy: 0.9426008968609866
Precision: 0.9574468085106383
Recall: 0.6

Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97       965
           1       0.96      0.60      0.74       150

    accuracy                           0.94      1115
   macro avg       0.95      0.80      0.85      1115
weighted avg       0.94      0.94      0.94      1115
```




```
After threshold adjustment to 0.4:
Precision: 0.9557522123893806
```